BALNE CHARITH CHANDRA SAI

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EDUCATION

University of Southern California (USC) | *Los Angeles* **|** *CA* **Aug 2022 – Dec 2023**

Master of Science in Computer Science

• Related Coursework: Advanced Analysis of Algorithms, Data Bases, Web Technologies, Advanced NLP

Gokaraju Rangaraju Institute of Engineering and Technology | *Hyderabad* **|** *TG* **Aug 2018 – May 2022**

Bachelor of Technology in Computer Science and Engineering

TECHNICAL SKILLS

Languages: Python, SQL, HTML, C, JavaScript, C++, Java, Swift.
Libraries: Pandas, NumPy, Scikit-Learn, Keras, TensorFlow, Matplotlib, NLTK, Spacy, ggplot2, Beautiful Soup, CV2.
Modeling: Statistical Modeling, Machine Learning, NLP, Deep Learning, Hypothesis Testing, A/B testing.
Technologies: MS-EXCEL, Flask, Git, FastApi, AWS, Google Cloud, Docker,Hive, Web Scraping, Hadoop.
Applied Skills: Mobile Application development, Full stack web development, problem-solving, Agile methodologies, Communication skills, Code optimization, Software development.

PROFESSIONAL EXPERIENCE

Software Engineer | *Web & mobile app development* **|** *USC Labs* **Mar 2023 - Present**

• Event Search: Developed web applications using Angular and Bootstrap as front-end technologies, with expertise in Node.js for making API calls to external services.

• Converted this web application into an iOS application while retaining the same back-end call functionality.

Software Engineer | *High-Performance Computing* **|** *Rivach Tech* **Mar 2020 - Aug 2021**

• Implemented machine learning modeling on commodity hardware, including general-purpose graphical processing units (GPUs) and multiple GPUs connected with message passing interface (MPI).

•Developed risk models involving iterative computations for credit account default probabilities using Markov Chain analysis.

• Conducted benchmarking to assess scalability and performance, demonstrating the superiority of single GPU training over CPU, multiple GPUs, and GPU-MPI combinations.

• Successfully applied the implemented approach to real-world data, showcasing the outperformance of single GPU training compared to CPU, Multiple GPU, and GPU-MPI configurations.

RESEARCH EXPERIENCE

Graduate Research Assistant | *University of Southern California* **Aug 2022 - Jan 2023**

• Collaborated with a team of senior researchers to develop a novel approach for generating synthetic cancer cells and detecting spike cancer cells within a blood sample.

• Utilized Wasserstein Generative Adversarial Networks (WGANs) to train a generative model using TensorFlow on USC's Mars supercomputer.

• Optimized code implementation, significantly reducing computation time from 8 hours to 3.5 hours, improving overall efficiency and productivity.

Research Assistant |*GRIET* **Oct 2020 – Jan 2022**

• Contributed to the advancement of healthcare technology by developing an innovative application for Covid-19 impact identification, combining questionnaire responses and cough analysis for effective assessment and monitoring.

• Implemented data augmentation techniques to generate a substantial 50GB dataset from an initial 10GB dataset, enhancing the diversity and richness of the training data.

• Parallel processed Mixed-data Neural Networks over 8-cores and achieved a high accuracy rate of 85% by leveraging the power of Multi-threading.

PROJECTS

Deep Learning-Based Mixed Data Neural Networks for Covid-19 Detection (*IEEE INDICON 2022 Publication*)

• Implemented manual feature extraction and mixed data feed-forward neural networks on cough sound to identify whether a person tested positive for covid-19.

• Extracted Mel-frequency coefficient features of cough sound samples and applied CNN to integrate with the ultimate pen layer of the multi-layer perceptron. Yielded Area Under Curve (AUC) score of 94% and Accuracy of 85%.

Geo-Spatial 3D Data Mapping for Drone Automation | *Python, CV, Deep-Learning*

• Implemented Convex Hull and found the nearest neighbors for the data points. Visualized data using Open Layers and created a Spirograph path for external path tracking.

• Deployed data to the drone's Autopilot module to track its optimized path using the Bellman-Ford algorithm.

ACHIEVEMENTS / EXTRA-CURRICULAR

• Google Certified Professional Data Engineer

• Participated and led a team of 5 in IEEE INDICON 2022 Conference challenge and bagged first place.

• Secured All India Scholars award in 2015 for Honors in matriculation.